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FORA Communication Protocol for SPO2 series

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Document Version:1.05

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Table Of Contents

1. GENERAL	
1.1. OVERVIEW	
1.2. DOCUMENT OVERVIEW	
1.3. DEFINITIONS , ACRONYMS , AND ABBREVIATIONS	
2. INTERFACE SPECIFICATION	
2.1. FRAME STRUCTURE	
2.2. COMMAND LIST	
2.2.1. [0 X23] READ DEVICE CLOCK TIME (4 BYTES)	
2.2.2. [0 X24] READ DEVICE MODEL	
2.2.3. [0 X25] READ THE STORAGE DATA WITH INDEX , PART 1(TIME)	
2.2.4. [0 X26] READ THE STORAGE DATA WITH INDEX , PART 2(RESULT)	
2.2.5. [0 X27] READ DEVICE SERIAL NUMBER , PART 1	

2.2.6.	[0 X28]	READ DEVICE SERIAL NUMBER , PART 2	
2.2.7.	[0 X2B]	READ STORAGE NUMBER OF DATA	
2.2.8.	[0 X33]	WRITE SYSTEM CLOCK TIME (4 BYTES)	
2.2.9.	[0 X49]	READ REAL TIME DATA	
2.2.10.	[0 X50]	TURN OFF THE DEVICE	
2.2.11.	[0 X52]	CLEAR /D ELETE ALL MEMORY	

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<http://foracare.com> SPO2 Meter Update:2015/11/06

Document Version:1.05

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1. General

1.1. Overview

The document describes the commands which will be u sed to communicate with spo2 meter.

Before retrieving data in the device, please make s ure you have already established the proper connect ion channel between the device and the host side. For e xample, if you use USB cable, be sure that the USB driver has been properly installed; or if you use B luetooth communication, be sure that the connection has been successfully established.

Generally speaking, the device will be the slave si de waiting for the request from the host which is u sually the computer or the hub/gateway.

Note for different Bluetooth versions:

For Bluetooth v2 meter: Please connect to meter by SPP (Serial Port Profile). Meter always stands by f or both pairing and connection, the PIN for paring is 111111.

For Bluetooth v3 meter: Please connect to meter by SPP (Serial Port Profile). Meter must be paired bef ore any connection, please find the instruction in mete r manual for the pairing process. The meter support s SSP (Simple Secure Pairing), as a result PIN is not required. If host does not support SSP, please use 0000 as PIN if required.

For Bluetooth v4 meter: Please connect to meter by following GATT service:

UUID Base: 1212-efde-1523-785feabcd123

Service UUID: 0x1523

Characteristic: 0x1524 (write/notify)

1.2. Document Overview

For the first time user on the device integration, please read the 2.1 frame structure. If you are fam iliar with serial port communication, you can review the comma nd list table in section 2.2 directly. The followin g sub sections are all about the details for each command .

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Document Version:1.05

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1.3. Definitions, Acronyms, and Abbreviations

GW Gateway / Hub / Computer side

MD SPO2 meter.
CMD Command. Usually its a one byte alphanumeric symbol.
ACK Acknowledgement. Usually its a one byte alphanumeric symbol.
NA Not available

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Document Version:1.05

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2. Interface specification
Block Diagram
The communication interface is base on RS232 serial port communication :

2.1. Frame Structure

Byte 1 2 3 ~ n-2 n-1 n
Name Start CMD Index, Address, or Data Stop Check
Sum
Format HEX
GW MD 51 CMD A3 [1n-1]
GW MD 51 ACK A5 [1n-1]
ie. Check sum means the summation from byte 1 to byte n-1 with 8-bit length

Example : 8-byte frame

Byte 1 2 3 4 5 6 7 8
Name Start CMD Data_0 Data_1 Data_2 Data_3 Stop ChkSum Baud rate 19200
Data bits 8
Parity No
Start bit 1
Stop bit 1 Medical Device /
Sensor
(MD) Gateway / Hub /
Computer
(GW) TX
RX
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Document Version:1.05

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Format HEX
GW MD 51 CMD Data_0 Data_1 Data_2 Data_3 A3 [1..7]
GW MD 51 ACK Data_0 Data_1 Data_2 Data_3 A5 [1..7]

ie. Check sum means thefrom byte 1 to byte 7 with 8 -bit length
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Document Version:1.05

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2.2. Command List

Message name	CMD/ACK	Direction	Length (Bytes)	Note
Read device clock time	0x23	GW MD	8	
	0x23	GW MD	8	
Read device model.	0x24	GW MD	8	
	0x24	GW MD	8	
Read storage data, part 1(time)	0x25	GW MD	8	
	0x25	GW MD	8	
Read storage data , part 2(result)	0x26	GW MD	8	
	0x26	GW MD	8	
Read device serial number, part 1	0x27	GW MD	8	
	0x27	GW MD	8	
Read device serial number, part 2	0x28	GW MD	8	
	0x28	GW MD	8	
Read storage number of data	0x2B	GW MD	8	
	0x2B	GW MD	8	
Write device clock time	0x33	GW MD	8	
	0x33	GW MD	8	
Read real time data	0x49	GW MD	8	
	0x49	GW MD	8	
Force Slave Device goes to Sleep (Turn OFF)	0x50	GW MD	8	
	0x50	GW MD	8	
Clear/Delete all memory	0x52	GW MD	8	
	0x52	GW MD	8	

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Document Version:1.05

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2.2.1. [0x23] Read device clock time (4 bytes)
Message Name: Read device clock time (4 bytes)
Message ID: 0x23
Message Description: Read device date and time (to minute)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read System Clock time	GW MD	0x23	0x0	0x0	0x0	0x0

Response from device. GW MD 0x23 Day + Month + Year Minute Hour

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x23 GW commands MD to response device clock time

DATA 4 0x0 NA

Field Size

(Bytes) Value Description.

ACK 1 0x23 MD response device clock time

DATA 2 Data_1 + Data_0 : Day+Month+Year (word) (see Table A)

DATA 2 Data_2 : Minute, Data_3 : Hour (see Table B)

Table A: Day + Month + Year

MSB LSB

Data_1 Data_0

Year (7-bit) Month (4-bit) Day (5-bit)

6 5 4 3 2 1 0 3 2 1 0 4 3 2 1 0

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Document Version:1.05

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Table B: Minute and Hour

2.2.2. [0x24] Read device model

Message Name: Read device model

Message ID: 0x24

Message Description: Read device model / project code

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read System Clock time GW MD 0x24 0x0 0x0 0x0 0x0

Response from device. GW MD 0x24 Project Code

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x24 GW commands MD to response device model

DATA 4 0x0

Field Size

(Bytes) Value Description.

ACK 1 0x24 MD response project code

DATA 2 Data_1 + Data_0 : Device model in four digits (Data_1 is the MSB LSB

Data_3 Data_2

NA Hour (5-bit) NA Minute (6-bit)

4 3 2 1 0 5 4 3 2 1 0

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Document Version:1.05

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higher byte)

DATA 2 Data_3, Data_2 : undefined

2.2.3. [0x25] Read the storage data with index, part 1(time)

Message Name: Read the storage data with index, part 1(time). Not all models are supported. PO100 is not supported.

Message ID: 0x25

Message Description: Read data (part 1: measure date and time)

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read System Clock time GW MD 0x2 5 Index 0x0 0x0

Response from device. GW MD 0x25 D_M_Year Minute Hour

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x25 GW commands MD to response measure data (part 1)

Index 2 0x00 is the latest reading in the device

Field Size

(Bytes) Value Description.

ACK 1 0x25 MD response measure date and time

D_M_Year 2 The measure date (see Table A in 2.2.1 [0x23])

Minute 1 The measure minute (see Table B)

Hour 1 The measure hour (see Table B)

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Document Version:1.05

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2.2.4. [0x26] Read the storage data with index, part 2(result)

Message Name: Read the storage data with index, part 2(result). Not all models are supported. PO100 is not supported.

Message ID: 0x26

Message Description: Read data (part 2: measure result)

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read System Clock time GW MD 0x26 Index 0x0 0x0
Response from device. GW MD 0x26 SpO2 0x0 Heart Rate

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x26 GW commands MD to response measure data (part 1)

Index 2 0x00 is the latest reading in the device

Field Size

(Bytes) Value Description.

ACK 1 0x26 MD response measure date and time

SpO2 2 The measure result (see Table C below)

The unit is %

Heart Rate 1 The measure result (see Table C below)

The unit is BMP(pul/mi n)

Table C: SpO2

MSB LSB

Data_1 Data_0

SpO2

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Document Version:1.05

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Table D: Heart rate

2.2.5. [0x27] Read device serial number, part 1

Message Name: Read the device serial number, part 1

Message ID: 0x27

Message Description: Read the device serial number, part 1. This command cooperates with 0x28 to complete the serial number.

Src / Des : GW / MD

Length: Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + + SN_0
(example: 32502102 4013102A, the former 4 bytes come from 0x28 and the later come from 0x27)

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read device serial

number, part 1 GW MD 0x27 0x0 0x0 0x0 0x0

Response from device GW MD 0x27 SN_0 SN_1 SN_2 SN_3

Field Description

Field Size

(Bytes) Value Description. 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

MSB LSB

Data_3 Data_2
Pulse (8-bit) (Unit= pul/min) N/A
7 6 5 4 3 2 1 0 7 6 5 4 3 2 1 0
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Document Version:1.05

Proprietary and confidential. 13 / 18

CMD 1 0x27 GW commands MD to response the device serial number,
part 1
DATA 4 0x0

Field Size
(Bytes) Value Description.
ACK 1 0x27 Read the device serial number, part 1
DATA 4 Data_0 ~ Data_3 : SN_0 ~ SN_3

2.2.6. [0x28] Read device serial number, part 2
Message Name: Read the device serial number , part 2
Message ID: 0x28
Message Description: Read the device serial number, part 2. This command cooperates with 0x27 to complete the serial number.
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + + SN_0
(example: 32502102 4013102A, the former 4 bytes come from 0x28 and the later come from 0x27)

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read device serial
number, part 2 GW MD 0x28 0x0 0x0 0x0 0x0
Response from device GW MD 0x28 SN_4 SN_5 SN_6 SN_7

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x28 GW commands MD to response the device serial number,
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Document Version:1.05

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part 2
DATA 4 0x0

Field Size
(Bytes) Value Description.
ACK 1 0x28 Read the device serial number , part 2
DATA 4 Data_0 ~ Data_3 : SN_4 ~ SN_7

2.2.7. [0x2B] Read storage number of data

Message Name: Read storage number of data. Not all models are supported. PO100 is not supported.

Message ID: 0x2B

Message Description: Read data counts

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read Storage Number of

Stored Readings GW MD 0x2B User 0x0 0x0 0x0

Response from device GW MD 0x2B Storage Number 0x0 0x0

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x2B GW commands MD to response the Number of Store d Readings

Field Size Value Description.

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Document Version:1.05

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(Bytes)

ACK 1 0x2B MD response the Number of Stored Readings

DATA 2 Data_1+Data_0 = Storage Number (Word)

2.2.8. [0x33] Write System Clock time (4 bytes)

Message Name: Write device clock time (4 bytes)

Message ID: 0x33

Message Description: Write the device date and time (to minute)

Src / Des: GW / MD

Length : Request: 8 bytes / Respons e: 8 bytes

Please refer to 0x23 command section for the format of date and time.

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Write System Clock time GW MD 0x33 Day+Month+Year Minute Hour

Response from device. GW MD 0x33 Day+Month+Yea r Minute Hour

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x33 GW commands MD to write System Clock time (4 b ytes)

DATA 2 Data_1 + Data_0 : Day+Month+Year (word)

DATA 2 Data_2 : Minute, Data_3 : Hour

Field Size
(Bytes) Value Description.
ACK 1 0x33 MD write System Clock time (4 bytes) O.K.
DATA 2 Data_1 + Data_0 : Day+Month+Year (word)
DATA 2 Data_2 : Minute, Data_3 : Hour
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Document Version:1.05

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2.2.9. [0x49] Read real time data
Message Name: Read real time data
Message ID: 0x49
Message Description: Read real time data
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read real time data GW MD 0x49 0x0 0x0 0x0 0x0
Response from device. GW MD 0x49 SpO2 0x0 Heart Rate

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x49 GW commands MD to response real time data
Data 4 0x00

Field Size
(Bytes) Value Description.
ACK 1 0x49 MD response real time measure data
SpO2 2 The measure result (see Table C)
The unit is %
Heart Rate 1 The measure result (see Table D)
The unit is BMP(pul/min)

2.2.10. [0x50] Turn off the device
Message Name: Turn off the device
Message ID: 0x 50
Message Description: Turn off the device from GW
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Document Version:1.05

Proprietary and confidential. 17 / 18

Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Turn off the device GW MD 0x50 0x0 0x0 0x0 0x0
GW MD 0x50 0x0 0x0 0x0 0x0

Field Description

Field Size
(Bytes) Value Description.
CMD 1 0x50 GW commands MD to turn off the power.
DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x50 MD responses.
DATA 4 0x0 NA

2.2.11. [0x52] Clear/Delete all Memory

Message Name: Clear/Delete all memory. Not all mode ls are supported. PO100 is not supported.

Message ID: 0x52

Message Description: Delete all measurement data in the device

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
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Document Version:1.05

Proprietary and confidential. 18 / 18

Clear/Delete all Memory
Storages GW MD 0x52 0x0 0x0 0x0 0x0
Response from device. GW MD 0x52 0x0 0x0 0x0 0x0

Field Description

Field Size
(Bytes) Value Description.
CMD 1 0x52 GW commands MD to Clear/Delete all Memory Stor ages
in Device EEPROM
DATA 1
DATA 3 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x52 Clear/Delete all Memory Storages in Device EEP ROM O.K.
DATA 1 User
DATA 3 0x0 NA